

UNIT 7 • GUIDED NOTES

Unit 7 — Data Analysis & *Statistics*

Interactive guided notes for Unit 7 — scatterplots, correlation, the line of best fit, and two-way tables. Fill in the blanks to learn — your progress saves automatically.

7.1

Scatterplots & Bivariate Data

NC.8.SP.1 — CONSTRUCT AND INTERPRET SCATTER PLOTS FOR BIVARIATE MEASUREMENT DATA.

What is bivariate data? Data that has *two* values for each item — like a person's height AND weight, or hours studied AND test score. We use a scatterplot to look for patterns between the two variables.

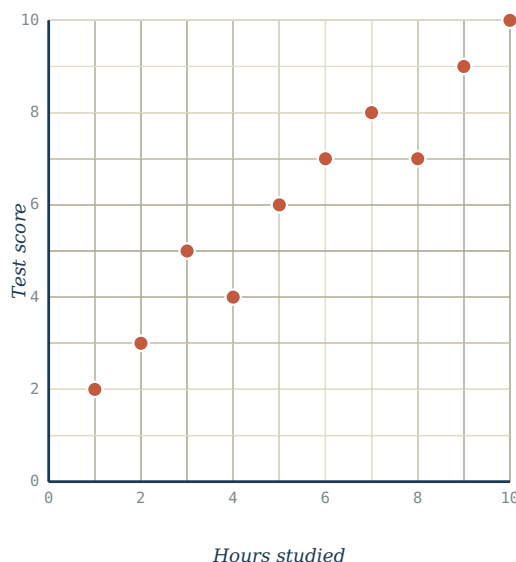
Univariate vs. Bivariate

Univariate data has variable per data point. Example: a list of test scores.

Bivariate data has variables per data point. Example: *hours studied* AND *test score* for each student.

 **The two variables together form an ordered pair (x, y) that we plot as a single dot.** Each dot is one person, one observation.

Anatomy of a Scatterplot



A scatterplot has:

An **x-axis** for the variable (the input).

A **y-axis** for the variable (the output).

Each dot represents data point (one person/item).

Both axes need a and a .

TRY IT

Practice: Suppose you collect data on 10 students: how many hours of sleep they got and their test score the next day.

What is the independent variable (x-axis)?

What is the dependent variable (y-axis)?

Each dot represents .

Total number of dots on the scatterplot:

💡 **Why scatterplots?** Bar graphs show frequencies of categories. Scatterplots show *relationships between two numerical quantities*. If you're asking "does X affect Y?", you want a scatterplot.

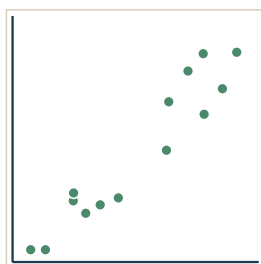
7.2

Correlation

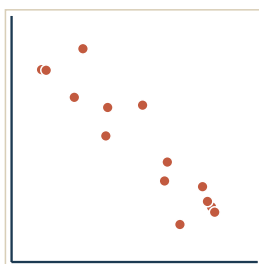
NC.8.SP.1 – DESCRIBE PATTERNS SUCH AS CLUSTERING, OUTLIERS, POSITIVE OR NEGATIVE ASSOCIATION, LINEAR ASSOCIATION, AND NONLINEAR ASSOCIATION.

What is correlation? Correlation describes how two variables tend to change together. There are three main types: *positive*, *negative*, and *none*. Plus two strengths: *strong* or *weak*.

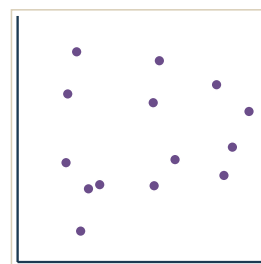
Three Types of Correlation



Positive



Negative



No Correlation

Positive correlation: as x increases, y also . The dots trend from left to right.

Negative correlation: as x increases, y [show](#) . The dots trend

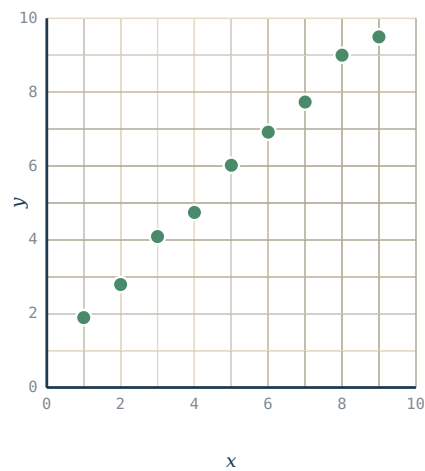
[show](#) from left to right.

No correlation: the dots are scattered with [show](#) clear pattern. As x increases, y does whatever it wants.

Strength of Correlation

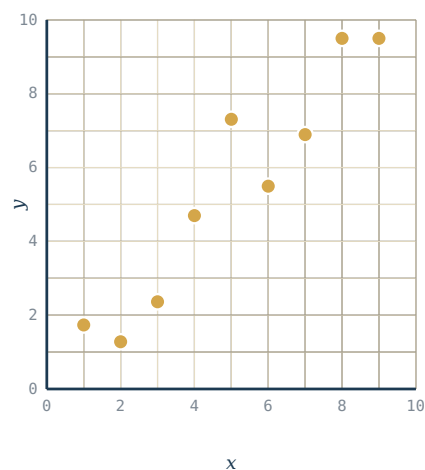
Even within "positive" or "negative", the correlation can be **strong** (dots close to a line) or **weak** (dots more scattered).

Strong positive



Dots tightly hug the trend line.

Weak positive



Dots are spread out around the trend.

TRY IT

Practice: For each scenario, predict the correlation type:

Hours studied vs. test score → [show](#)

Number of absences vs. final grade → [show](#)

Shoe size vs. IQ → [show](#)

Outdoor temperature vs. ice cream sales → [show](#)

Age of a car vs. value of the car → [show](#)

⚠ **Correlation $\neq$ Causation:** Just because two things are correlated doesn't mean one *causes* the other! Ice cream sales and shark attacks are both higher in summer — but ice cream doesn't cause sharks. Both are caused by hot weather (summer).

🔍 **Outliers and clusters:** An *outlier* is a dot that's far from the rest. A *cluster* is a group of dots tightly grouped together. Both are worth pointing out when you describe a scatterplot.

7.3

Line of Best Fit

NC.8.SP.2 / NC.8.SP.3 – INFORMALLY FIT A STRAIGHT LINE FOR SCATTER PLOTS THAT SUGGEST A LINEAR ASSOCIATION; USE THE EQUATION TO SOLVE PROBLEMS AND MAKE PREDICTIONS.

What is a line of best fit? When the dots in a scatterplot suggest a linear pattern, we can draw a single straight line that goes through the "middle" of the cloud of dots. This line represents the overall trend and helps us make predictions.

When Should You Draw One?

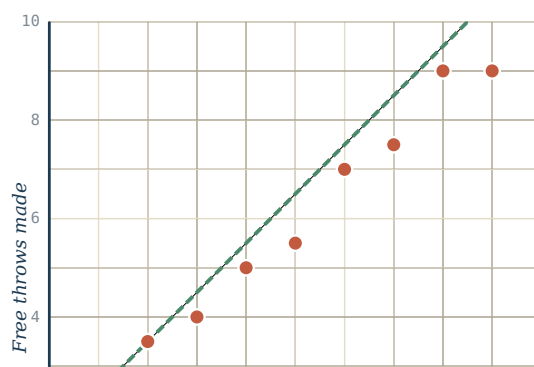
A line of best fit only makes sense when:

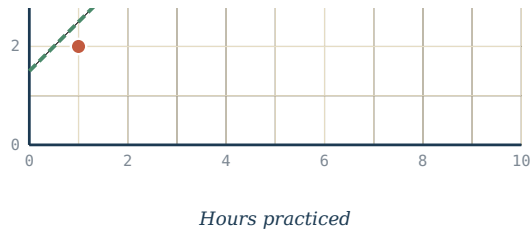
The data shows a [show](#) pattern (positive or negative correlation).

You should NOT draw one if the dots show [show](#) correlation or a curved/non-linear pattern.

EXAMPLE

Example: The scatterplot below shows hours practiced vs. free throws made for a basketball team. The dashed green line is the line of best fit.





Notice the line:

- Passes through the "middle" of the dots
- Has roughly the same number of dots above and below it
- Follows the overall positive trend
- Doesn't have to touch any specific dot

Drawing a Line of Best Fit (Informally)

1. Look at the overall direction of the dots (positive or negative).
2. Use a ruler — try to put the line so dots are roughly balanced above and below.
3. Don't connect the dots! It's a smooth straight line, not a zigzag.
4. Don't be tricked by outliers — ignore dots far from the rest.

Using the Line to Make Predictions

Once you draw a line of best fit, you can:

- Find its equation ($y = mx + b$) by reading off the slope and y-intercept.
- Plug in a new x-value to predict y.
- Read graphically: find x on the x-axis, go up to the line, then over to the y-axis.

TRY IT

Practice: A line of best fit for a scatterplot has equation $y = 2x + 1$ where x is hours studied and y is test score.

The slope means: each extra hour of study adds about [show](#) points to the test score.

The y-intercept means: with 0 hours of study, the predicted score is [show](#) .

If a student studies 4 hours, predicted score = $2(4) + 1 =$ [show](#) .

If a student studies 8 hours, predicted score = [show](#) .

⚠ **Predictions are approximate.** The line is a model, not exact. A student who studies 4 hours might get 7 or 11 — the line predicts about 9 on average.

Two-Way Tables

NC.8.SP.4 – UNDERSTAND PATTERNS OF ASSOCIATION BETWEEN TWO VARIABLES BY DISPLAYING FREQUENCIES AND RELATIVE FREQUENCIES IN A TWO-WAY TABLE.

When do we use them? Two-way tables help organize *categorical* bivariate data — when both variables are categories instead of numbers. Example: grade level and lunch preference, gender and favorite sport.

Reading a Two-Way Table

A school surveyed 120 students about whether they like pizza, sorted by grade:

	Like Pizza	Don't Like Pizza	Total
6th Grade	32	8	40
7th Grade	28	12	40
8th Grade	24	16	40
Total	84	36	120

Total students surveyed:

Number of 6th graders who like pizza:

Total 7th graders surveyed:

Total students who don't like pizza:

Relative Frequency

Relative frequency = the count divided by a total. Same idea as in Unit 6 frequency tables, but now we have multiple totals to choose from.

The fraction of *all* 120 students who like pizza: $84 \div 120 =$ or

%

The fraction of *6th graders* who like pizza: $32 \div 40 =$ or

%

The fraction of *8th graders* who like pizza: $24 \div 40 =$ or

%

 **Looking for an association:** Compare the relative frequencies. Above, 80% of 6th graders like pizza but only 60% of 8th graders do. There's an *association* between grade level and pizza preference — they're related!

TRY IT

Practice: A teacher surveys 100 students about whether they have a pet.

	<i>Has pet</i>	<i>No pet</i>	<i>Total</i>
Boys	20	30	50
Girls	35	15	50
Total	55	45	100

Total students surveyed:

Fraction of all students who have a pet: or %

Fraction of *boys* who have a pet: or %

Fraction of *girls* who have a pet: or %

Is there an association between gender and pet ownership? (girls are much more likely to have one in this survey)



Review & Error Analysis

SPOT THE COMMON UNIT 7 MISTAKES – CORRELATION VS. CAUSATION, THE LINE OF BEST FIT, AND READING TWO-WAY TABLES FAIRLY.

🔍 Spot the Error: Correlation vs. Causation

✗ **Student says:** "Ice cream sales and drowning rates have a positive correlation, so eating ice cream causes drowning."

✓ **Fix:** Correlation does NOT mean causation! Both increase in summer because of a **lurking variable** — hot weather. Ice cream doesn't cause drowning!

💡 *Always look for a third variable that could explain the relationship. Two things trending together doesn't mean one causes the other.*

🔍 Spot the Error: Line of Best Fit

✗ **Student says:** "The line of best fit must pass through every data point."

✓ **Fix:** The line of best fit shows the **general trend**. It should have roughly equal numbers of points above and below it — it does NOT need to pass through any specific points.

🔍 Spot the Error: Reading a Two-Way Table

✗ **Student says:** "More 8th graders like basketball (35) than 7th graders (30), so 8th graders are more into basketball." But 8th grade has 80 students total vs 70 in 7th grade!

✓ **Fix:** Use **relative frequency** to compare fairly: 7th = $30/70 \approx 42.9\%$ vs 8th = $35/80 \approx 43.75\%$. They're nearly the same! Raw counts can be misleading when group sizes differ.

💡 *When comparing groups of different sizes, ALWAYS use percentages (relative frequency), not raw counts.*